$$x = x_0 + v_0 + \frac{1}{2}at^2$$
$$v = v_0 + at$$
$$v^2 = v_0^2 + 2a(x - x_0)$$
$$Galilean: t' = t, x' = x - vt$$
$$Lorentz: ct' = \gamma ct - \gamma\beta x$$
$$x' = \gamma x - \gamma\beta ct$$
$$Inv\ Lorentz: ct = \gamma ct' + \gamma\beta x'$$
$$x = \gamma x' + \gamma\beta ct'$$

$$(\Delta s)^2 = (\Delta ct)^2 - (\Delta x)^2 = (\Delta ct')^2 - (\Delta x')^2 = (\Delta s')^2$$

$$v = \frac{v' + \beta c}{c + \beta v'}c,$$
$$Vel\ v' in\ rest\ frame\ and\ v\ in\ stationary\ frame$$
$$KE = mc^2(\gamma - 1)$$
$$E_{Total} = KE + mc^2 = \gamma mc^2$$
$$m_{Relativistic} = \gamma m$$

$$v = \lambda f = \lambda v = c$$
$$\theta_i = \theta_r$$
$$\theta_c = arcSin(\frac{n_1}{n_2})$$
$$f = \frac{n_2 r}{n_2 - n_1}$$
$$\frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}$$
$$m' = \frac{h_i}{h_o} = \frac{d_i}{d_o},$$
$$2t = \left(m + \frac{s}{2}\right)\lambda, Constructive\ reflection\ (s\ phase\ shifts)$$
$$Sin(\theta) = \frac{m\lambda}{a}, Single\ Slit\ (minima)$$
$$Sin(\theta) = \frac{m\lambda}{d}, Aperture\ (minima)$$
$$I = I_{Max}\left(Cos(\beta)\right)^2\left(\frac{Sin(\alpha)}{\alpha}\right)^2, Multislit, b\ is\ slit\ spacing$$
$$m\lambda = dSin(\theta), Maxima$$

$$\gamma = \frac{\delta}{2}, \alpha = \frac{\pi a}{\lambda}Sin(\theta), \delta = \frac{2\pi}{\lambda}\Delta r = \frac{2\pi d}{\lambda}Sin(\theta)$$
$$x = \frac{lm\lambda}{\sqrt{b^2 - m^2\lambda^2}}, m^{th} Bright\ Two\ slit$$

$$c = 2.998 \times 10^8 ms^{-1}$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$$
$$\omega = \omega_0 + \alpha t$$
$$\omega^2 = \omega^2 + 2\alpha(\theta - \theta_0)$$

$$\beta = \frac{v}{c}, \gamma = \frac{1}{\sqrt{1 - \beta^2}}$$
$$\Delta t = \gamma \Delta t_{Proper}, t_{Proper} is\ shortest$$
$$\Delta x = \frac{1}{\gamma}\Delta x_{Proper}, x_{proper}\ is\ longest$$
$$\Delta s^2 < 0 - Spacelike$$
$$\Delta s^2 = 0 - light\ like$$
$$\Delta s^2 > 0 - Time\ like$$
$$v' = \frac{v - \beta c}{c - \beta v}c,$$
$$Vel\ v\ in\ rest\ frame\ and\ v'\ in\ moving\ frame$$
$$p = \gamma mv = \sqrt{\gamma^2 - 1}mc$$
$$\therefore E^2 = (pc)^2 + (mc^2)^2$$
$$n = \frac{c}{v}$$
$$E_{Total} = E_{Absorbed} + E_{Reflected} + E_{Transmitted}$$
$$n_1 Sin(\theta_1) = n_2 Sin(\theta_2)$$
$$f_{Sphr\ Mirr} = \frac{r}{2}$$
$$\frac{n_1}{d_0} + \frac{n_2}{d_i} = \frac{n_2 - n_1}{r}$$
$$\frac{1}{f} = (n_{12} - 1)\left(\frac{1}{r_1} - \frac{1}{r_2}\right), (n_{12} = \frac{n_2}{n_1})$$
$$\phi = \pi: n_1 < n_2, \phi = 0: n_1 > n_2$$

$$2t = \left(m + \frac{s+1}{2}\right)\lambda, Destructive\ reflection$$
$$I = I_{Max}\left(\frac{Sin(\alpha)}{\alpha}\right)^2, \alpha = \frac{\phi}{2} = \frac{\pi a}{\lambda}Sin(\theta)$$
$$\theta_c = 1.22\frac{\lambda}{d}$$
$$\beta = \frac{\pi b}{\lambda}Sin(\theta), \alpha = \frac{\pi a}{\lambda}Sin(\theta)$$
$$I = I_0\left(\frac{Sin(\alpha)}{\alpha}\right)^2\left(\frac{Sin(N\gamma)}{Sin(\gamma)}\right)^2$$

$$\frac{\lambda}{\Delta\lambda} = mN = \frac{mw}{d}$$
$$x = \frac{l(m - \frac{1}{2})\lambda}{\sqrt{b^2 - (m - \frac{1}{2})^2\lambda^2}}, m^{th} Dark\ Two\ slit$$

$$I_s(\lambda,T)d\lambda = \frac{2\pi hc^2}{\lambda^5}\frac{1}{e^{\frac{hc}{\lambda kT}}-1}d\lambda \;\; [Js^{-1}m^{-2}m^{-1}]$$

$$\lambda_{max}T = 2.898\times10^{-3}\,[mK]$$

$$E = hv$$

$$1.602\times10^{-19}\,[\frac{J}{eV}]$$

$$p = \frac{E}{c} = \frac{hv}{c} = \frac{h}{\lambda}$$

$$\frac{hc}{\lambda} = \frac{hc}{\lambda'} + K = \frac{hc}{\lambda'} + (\gamma-1)m_0c^2$$

$$\frac{h}{\lambda} = \frac{h}{\lambda'}Cos(\phi) + \gamma m_0 v Cos(\theta)\;\{x\}$$

$$0 = \frac{h}{\lambda'}Sin(\phi) - \gamma m_0 v Sin(\theta)\;\{y\}$$

$$k = \frac{2\pi}{\lambda} = \frac{p}{\hbar} = \frac{\sqrt{2mE}}{\hbar}\,(Angular\ Wavenumber)$$

$$T = \frac{16k^2b^2}{(b^2+k^2)^2}e^{-2bl} \approx \left[\frac{16E}{U}\left(1-\frac{E}{U_b}\right)\right]e^{-2bl}$$

$$\infty\ U\ Well: kl = n\pi, k = \frac{\sqrt{2mE}}{\hbar}$$

$$E_n = \left(\frac{\hbar^2\pi^2}{2ml^2}\right)n^2 = \left(\frac{h^2}{8ml^2}\right)n^2$$

$$L = n\hbar$$

$$E_n = -\left(\frac{m_e e^4}{8\epsilon_0^2 h^2}\right)\frac{1}{n^2} = -\frac{13.6eV}{n^2}$$

$$E_{n_x,n_y,n_z} = \frac{h^2}{8m}\left(\frac{n_x^2}{l_x^2}+\frac{n_y^2}{l_y^2}+\frac{n_z^2}{l_z^2}\right)$$

$$n - Principle\ QN\ (E_n), \ell - Orbital\ QN$$

$$m_\ell \in (-\ell,\ell) - Orbital\ Magnetic\ QN$$

$$s - Spin\ \ QN\ \left(\frac{1}{2}for\ pne\right), m_s = \frac{1}{2}, -\frac{1}{2} - spin\ QN$$

$$L = \sqrt{\ell(\ell+1)}\hbar$$

$$\mu_{orb} = -\frac{\ell}{2m}L = -\frac{\ell}{2m}\sqrt{\ell(\ell+1)}\hbar$$

$$\mu_{orb,z} = -\frac{\ell}{2m}L_z = -\frac{\ell}{2m}m_\ell\hbar = -m_\ell\mu_B$$

$$\mu_B = \frac{\ell\hbar}{2m} = 9.724\times10^{-24}JT^{-1}$$

$$For\ a\ given\ L\ there\ will\ be\ 2\ell+1\ L_z$$

$$Bremstrahlung: \frac{1}{\lambda_{K\alpha}} = \frac{3}{4}R(z-1)^2\left(\frac{1}{n_2^2}-\frac{1}{n_1^2}\right), z = Atomic\ Num$$

$$\Delta E_{ben} = \frac{\Delta E_{be}}{A}$$

$$\beta^-: {}^A_ZX \rightarrow {}^A_{z+1}Y + e^- + \bar{\nu} + E$$

$$\beta^+: {}^A_ZX \rightarrow {}^A_{z-1}Y + e^+ + \nu + E$$

$$R \equiv -\frac{dN}{dt} = \lambda N = R_0e^{-\lambda t}$$

$$I = \sigma T^4, 5.67\times10^{-8}, \hbar = \frac{h}{2\pi}$$

$$L = \sigma T^4 4\pi R^2$$

$$K_{max} = h\nu - \phi$$

$$\lambda = \frac{h}{p} = \frac{h}{\gamma m_0 v}$$

$$\Delta = \lambda' - \lambda = \frac{h}{m_0c}(1-Cos(\phi))$$

$$\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$$

φ = photon, θ = electron

$$|\psi|^2 = \left|\psi_0e^{-ikx}\right|^2 = \psi_0^2(e^{-ikx})(e^{-ikx})^* = \psi_0^2 = Const$$

$$\Delta x\Delta p \geq \frac{\hbar}{2}\ or\ \Delta E\Delta t \geq \frac{\hbar}{2}$$

$$Where\ b = \sqrt{\frac{2m(U_b-E)}{\hbar^2}}$$

$$\psi_n(x) = \sqrt{\frac{2}{l}}Sin\left(\frac{n\pi}{l}x\right)$$

$$\Delta E = E_{High} - E_{Low} = \frac{h^2}{8ml^2}(n_{High}^2 - n_{Low}^2)$$

$$r_n = \frac{\epsilon_0h^2}{\pi m_e e^2}n^2 = an^2, a = \ 5.29\times10^{-11}[m]$$

$$\frac{1}{\lambda} = R\left(\frac{1}{n_{Low}^2}-\frac{1}{n_{High}^2}\right), R = 1.097\times10^7[m^{-1}]$$

$$Orbitals: s,p,d,f,g \rightarrow \ell = 0,1,2,3,4,\cdots$$

$$\psi(r) = \frac{1}{\sqrt{\pi}}a^{-\frac{3}{2}}e^{-\frac{r}{a}}\,(H\ Gnd\ state)$$

$$\psi^2(r)dV = \frac{4}{a^3}e^{-\frac{2r}{a}}r^2dr$$

$1s$,
$2s\ 2p$,
$3s\ 3p\ 3d$,
$4s\ 4p\ 4d\ 4f$,
$5s\ 5p\ 5d\ 5f$,
$6s\ 6p\ 6d$,
$7s\ 7p$,
$2, 6, 10, 14$

$$S = \sqrt{s(s+1)}\hbar = \frac{\sqrt{3}}{2}\hbar, \mu_s = -\frac{\ell}{m}S$$

$$1u = 931.494013MeV/c^2$$

$$\Delta = M - \sum m$$

$$E_{be} = \sum(mc^2) - Mc^2 = (\Delta m)c^2$$

$$\alpha: {}^A_ZX \rightarrow {}^{A-4}_{Z-2}Y + {}^4_2He + E$$

$$\gamma: {}^A_ZX^* \rightarrow {}^A_ZX + \gamma + E$$

$$N = N_0e^{-\lambda t}, t_{\frac{1}{2}} = \frac{\ln(2)}{\lambda}$$